VEMULA SYAM PRAKASH

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PROFESSIONAL SUMMARY

Master of Computer Applications graduate with a Statistics background and hands-on project experience in AWS cloud infrastructure, data engineering, backend services, and applied AI. Designed and deployed ETL pipelines on S3/Glue/Athena, built distributed systems in Go and Python, created multi-agent AI tools with LangGraph, and delivered live monitored applications using Docker, Terraform, FastAPI, and Flask.

TECHNICAL SKILLS

Cloud / AWS: S3, Boto3, AWS CLI, EC2, Lambda, RDS, IAM, CloudWatch, Elastic Beanstalk, SageMaker, Glue, Athena
Infrastructure / DevOps: Docker, Terraform, GitHub Actions, Prometheus, Grafana, Render.com
Data Engineering: ETL Pipeline Architecture, Pandas, NumPy, PostgreSQL, MySQL, Redis, ClickHouse, SQLite
Programming: Python, SQL, Go, JavaScript
Backend / APIs: FastAPI, Flask, REST APIs, Gin
ML / AI: Scikit-learn, Random Forest, Gradient Boosting, SHAP, LangGraph, RAG, ChromaDB, Ollama
Visualization: Streamlit, Plotly, Matplotlib

PROJECTS

AWS E-Commerce Data Pipeline | Python, Pandas, Plotly, Streamlit, AWS S3, Glue, Athena **Sep 2026**
GitHub | Live Dashboard on Render

- Designed end-to-end ETL architecture (Raw S3 → AWS Glue → Curated S3 → Athena → Dashboard) for sales, category, and delivery data.
- Built Streamlit analytics dashboard with CSV upload/validation, revenue/order-volume/AOV KPIs, trend analysis, and late-delivery insights.
- Structured curated Athena tables by category, delivery, and sales dimensions for easy KPI extension; deployed as live web service.

Drug & Dosage Decision-Support Prototype | Python, Scikit-learn, Flask, AWS **Jul 2026**
GitHub | Live API on AWS | Springer-format Paper

- Developed ML prototype predicting drugs and dosages across 5 conditions and 15 drugs, achieving 94% accuracy on 941 patient records using Random Forest and Gradient Boosting with cross-validation and F1 evaluation.
- Added SHAP explainability and 5 rule-based safety overrides; deployed Flask REST API on Elastic Beanstalk, S3, RDS, SageMaker & CloudWatch with 5-screen UI and PDF reporting.

Grounded RAG — Document-Grounded RAG Application | Python, FastAPI, ChromaDB, Ollama **Jul 2026**
GitHub | Live Demo on Render

- Built production RAG system using FastAPI + ChromaDB for PDF/TXT/Markdown ingestion and source-grounded answers with citation support.
- Tuned chunking/retrieval, supported Ollama and OpenAI-compatible providers, added API-key protection and CI; deployed live on Render.

CodeForge — Autonomous Software Factory | Python, LangGraph, FastAPI, PostgreSQL, Docker **Jun 2026**
GitHub | Live on Render + GitHub Actions CI

- Built multi-agent system (Planner, Architect, Coder, Tester, Reviewer, Security) orchestrated with LangGraph, including autonomous test-debug-retest loop using AST ingestion and hybrid retrieval.
- Automated security checks, GitHub PR creation, and observability; deployed on Render with CI.

NanoLink — Distributed High-Throughput URL Shortener **Mar 2026**
GitHub | Live Service

- Developed distributed URL shortener with REST APIs for creation, redirection, and QR generation; implemented Redis caching/rate-limiting, PostgreSQL persistence, and ClickHouse analytics.
- Load-tested redirect path, containerized with Docker, provisioned with Terraform, and monitored with Prometheus/Grafana.

EDUCATION

Vignan's Lara Institute of Technology and Science, Vadlamudi — Jul 2026
Master of Computer Applications (MCA)
Acharya Nagarjuna University, Guntur — Jul 2024
Bachelor of Science (Statistics)

CERTIFICATIONS

- AWS Cloud Practitioner (CLF-C02) — AWS | 2026
- Google AI Professional Certificate — Google/Coursera | Aug 2026
- Oracle Certified Foundations Associate – Agentic AI — Oracle | 2026